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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE O Level

MARK SCHEME for the May/June 2006 question paper

5070 CHEMISTRY

5070/03 **Paper 3** **maximum raw mark 40**

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All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

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1 16 marks

(a) titration

4 marks for each of two titration results within 0.2 cm^3 of the Supervisor's value.
3 marks for results within 0.3 cm^3 etc.
no marks for results more than 0.5 cm^3 from the supervisor's value

maximum of 3 marks for concordance, i.e. results within 0.2 cm^3

1 mark for taking a correct average

(b) concentration of **X** **[2]**

method (1) answer (1)

(c) M_r of **X** **[2]**

method (1) answer (1)

2 **24 marks**

Solid R (manganese(IV) oxide)

Test 1	effervesces	**[1]**
	gas relights a glowing splint	**[1]**
	oxygen produced	**[1]**
	exothermic reaction	**[1]**
	solid is insoluble (or appears unchanged)	**[1]**

Test 2	effervesces	**[1]**
	gas bleaches litmus	**[1]**
	chlorine produced	**[1]**
	brown solution	**[1]**

Solution S (manganese(II) sulphate)

Test 3	beige ppt/pale brown	**[2]**
	insoluble in excess	**[1]**
	darkens on the surface	**[1]**
	brown ppt	**[1]**
	effervesces	**[1]**
	gas relights a glowing splint or oxygen produced	**[1]**

Test 4	no reaction	**[1]**
	no reaction	**[1]**

Test 5	white ppt	**[1]**
	insoluble in acid	**[1]**

Test 6	brown ppt	**[1]**

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Conclusion

In Test 1 R is a catalyst [

In Test 2 R is an oxidising agent **[1]**

The anion is a sulphate **[1]**

Any 24 marks to score.